

Jadavpur University



Computer Networks **Lab Report**

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YEAR : BCSE UG – III (A1)

SESSION : 2023 – 27

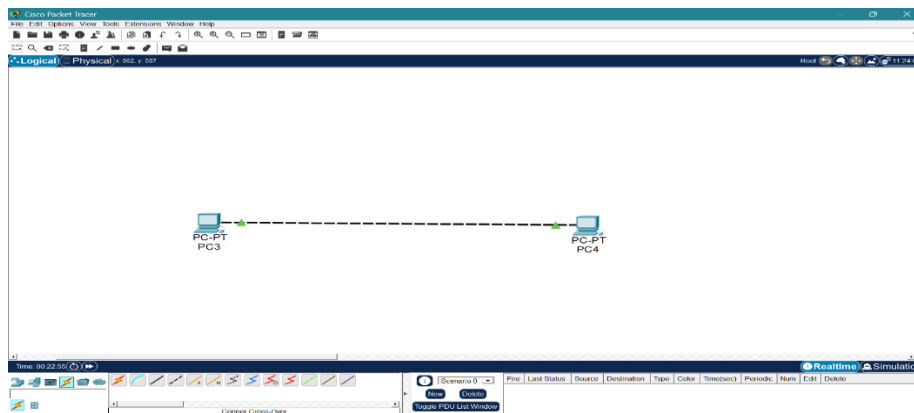
ASSIGNMENT : 0 6

PROBLEM STATEMENT: Use Cisco Packet Tracer software to do the following experiments.

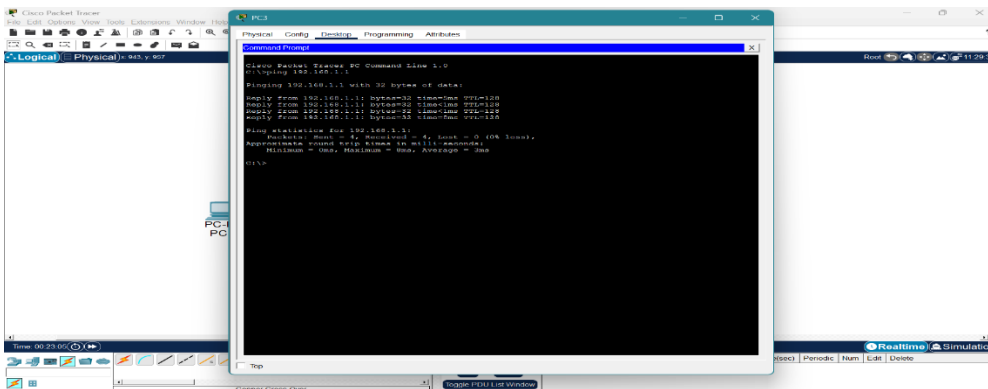
Overview : This entire assignment has been done using the CISCO Packet tracer tool.

System : Linux (Ubuntu 20.04)

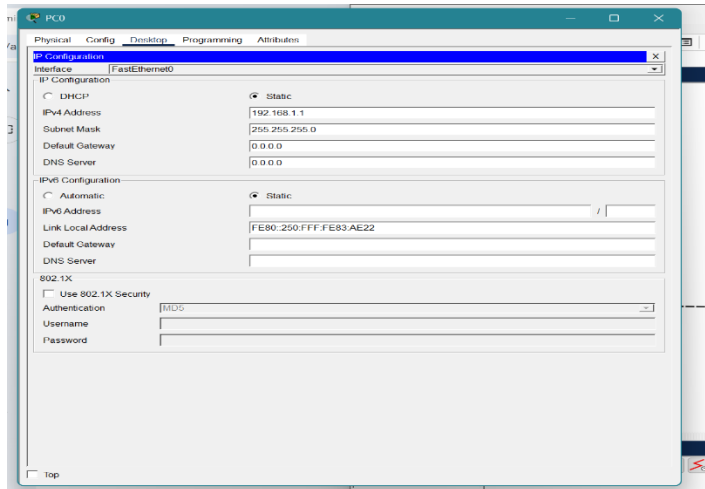
1. Connect two hosts back-to-back with a crossover cable. Assign IP addresses, and see whether they are able to ping each other.



Main Diagram

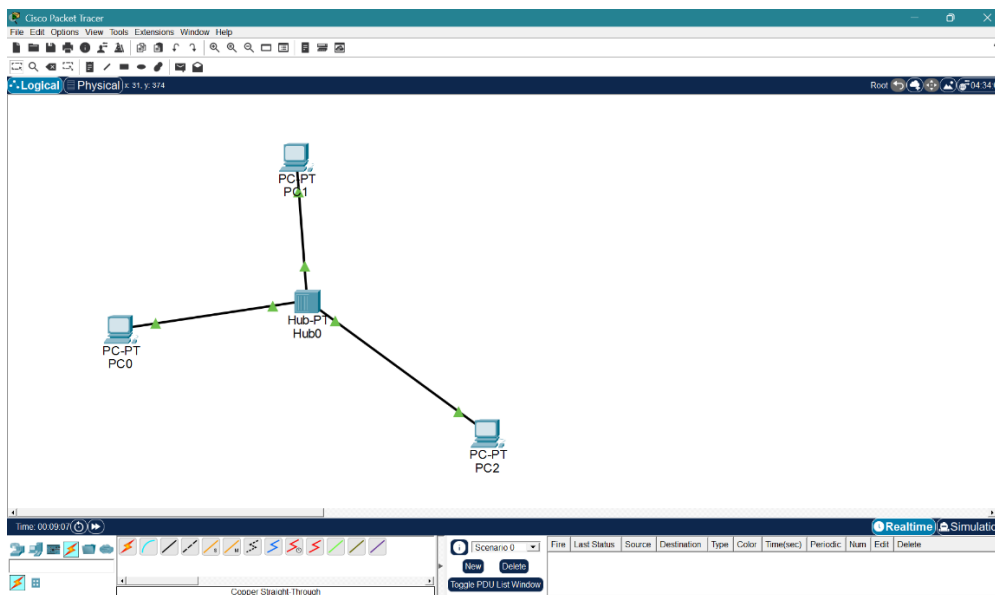


Output PC0

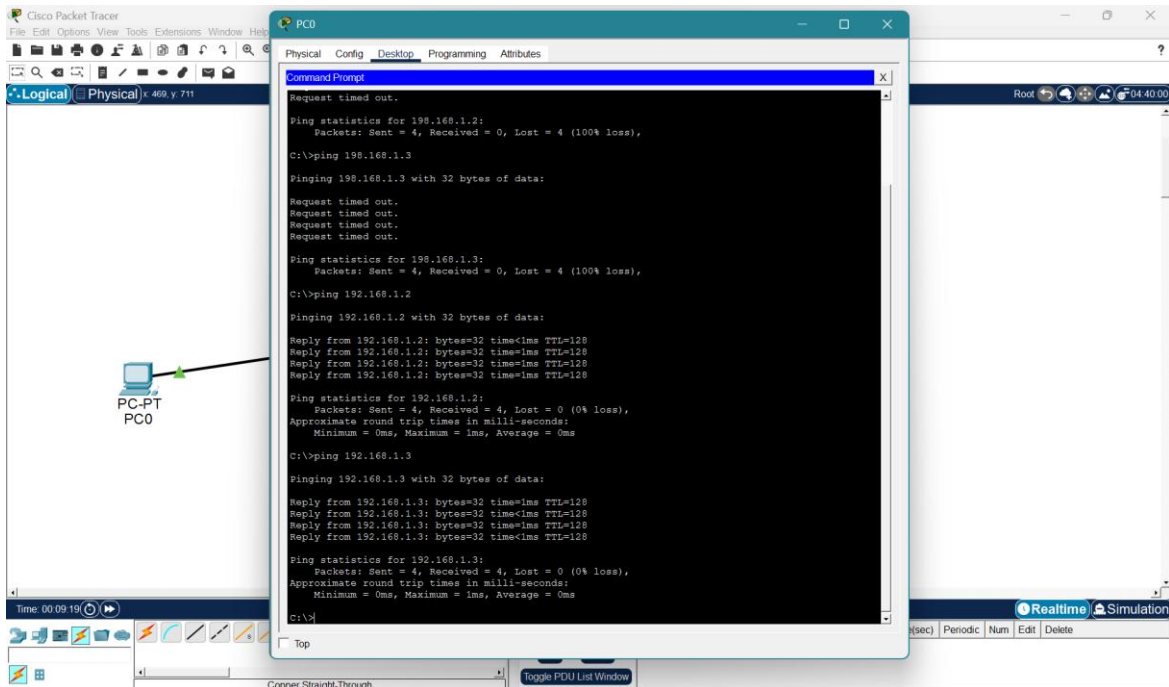


PC0 Config

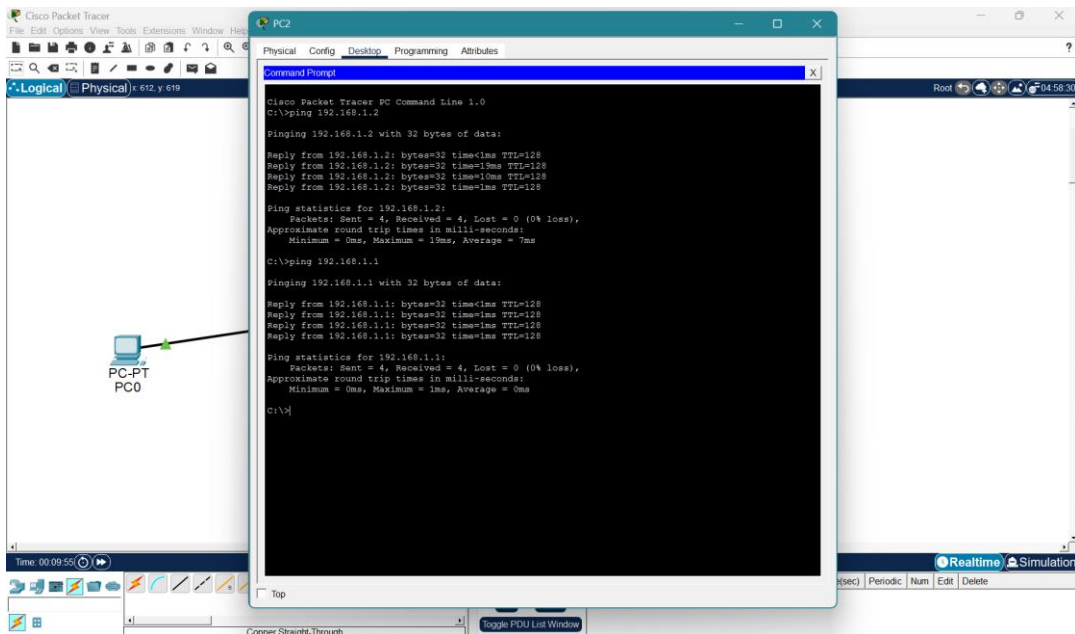
2. Create a LAN (named LAN-A) with 3 hosts using a hub. Ping each pair of nodes.



Main Diagram

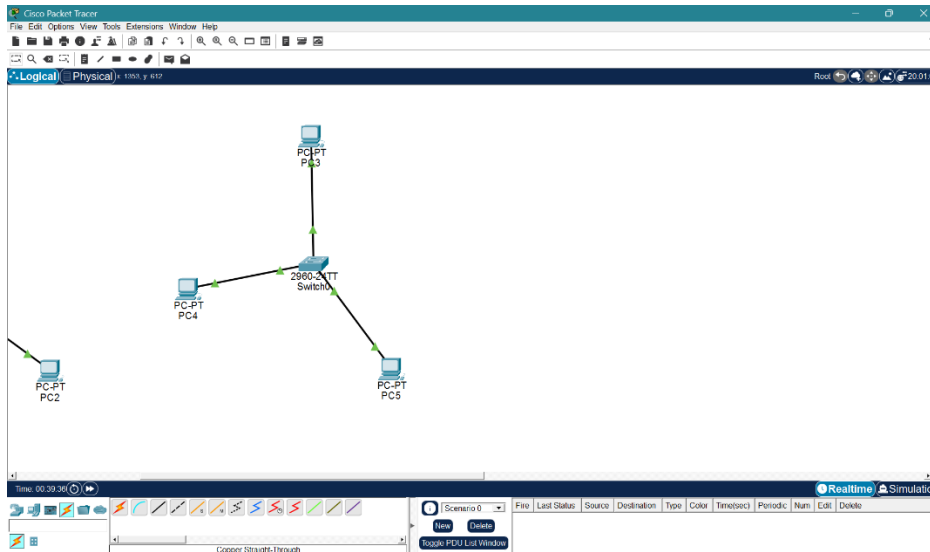


PC0 Ping

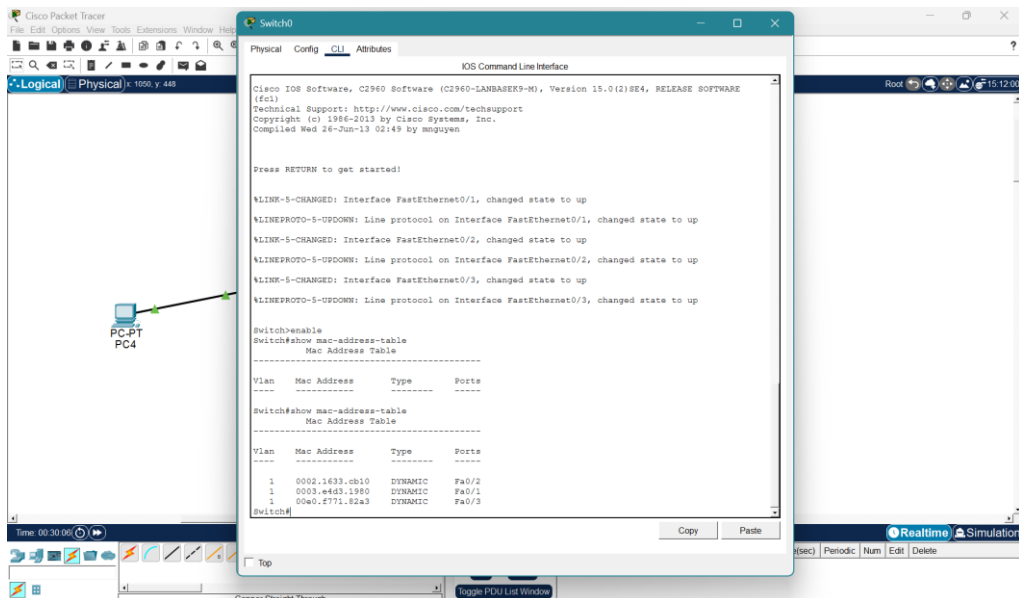


PC2 Ping

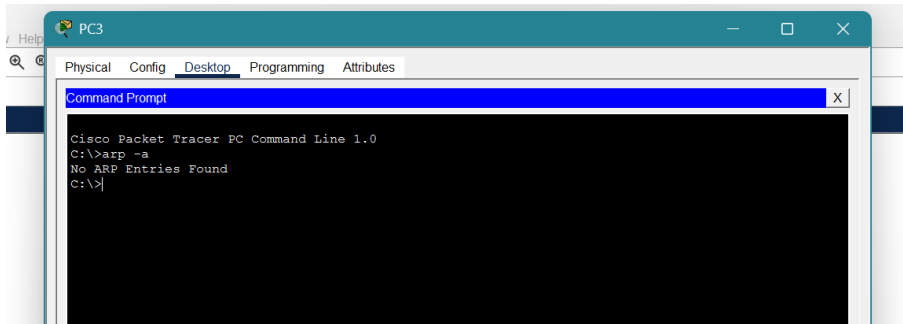
3. Create a LAN (named LAN-B) with 3 hosts using a switch. Record contents of the ARP Table of end hosts and the MAC Forwarding Table of the switch. Ping each pair of nodes. Now record the contents of the ARP Table of end hosts and the MAC Forwarding Table of the switch again.



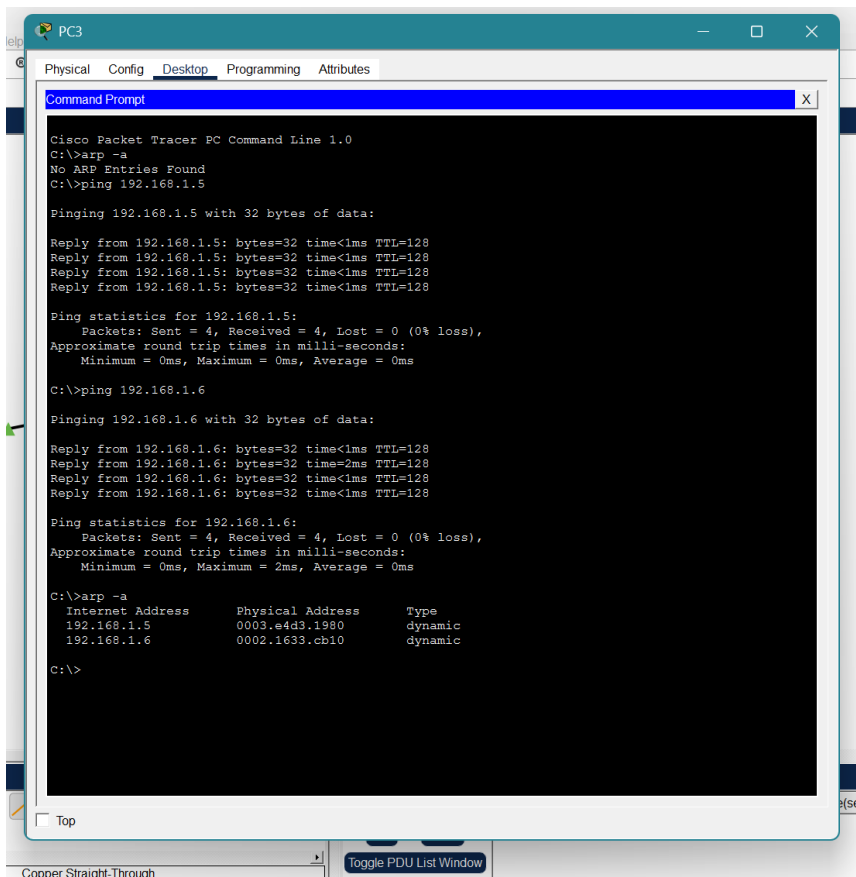
Main Diagram



MAC Address Table after Ping

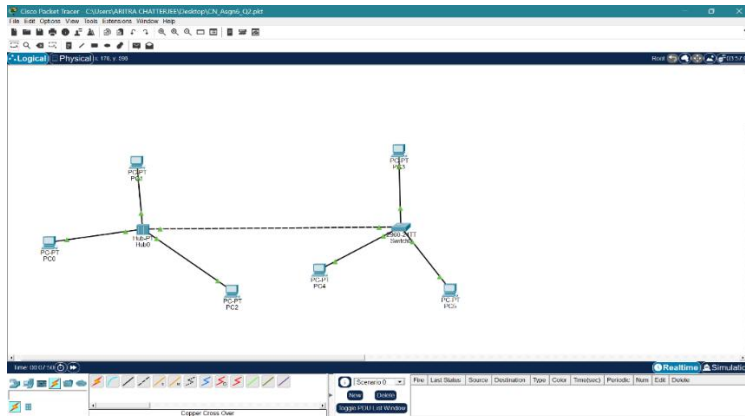


PC3 Before Ping

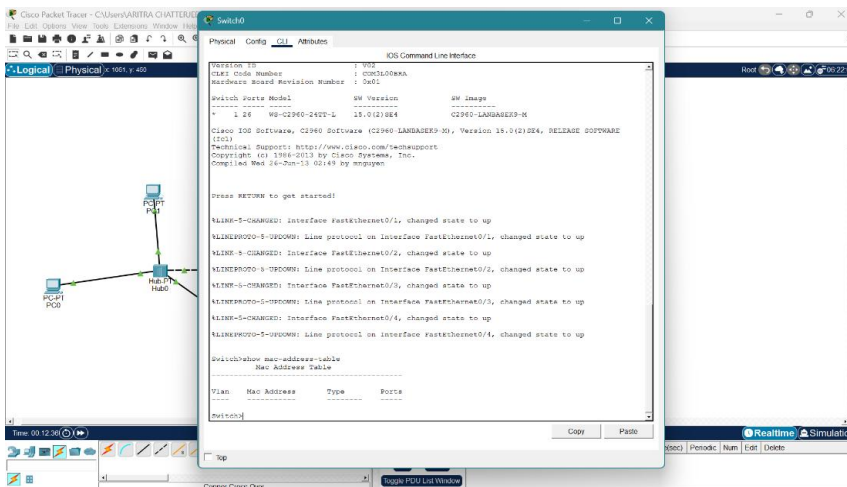


PC3 after Ping

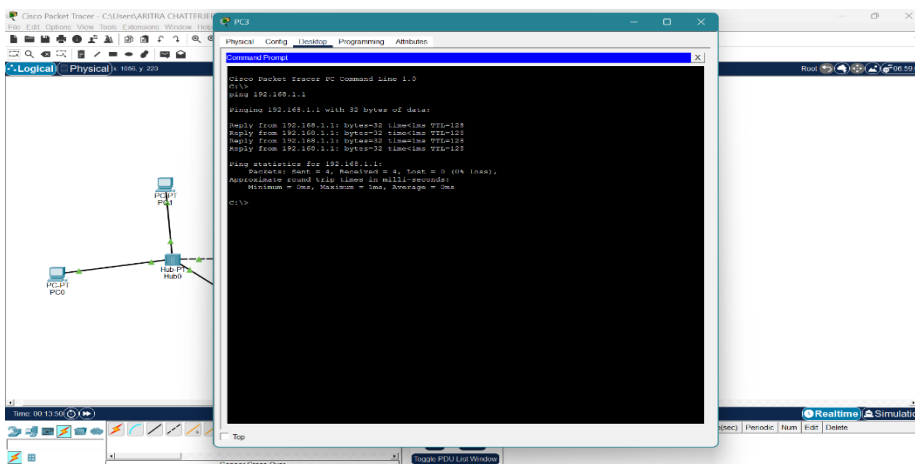
4. Connect LAN-A and LAN-B by connecting the hub and switch using a crossover cable. Ping between each pair of hosts of LAN-A and LAN-B. Now record the contents of the ARP Table of end hosts and the MAC Forwarding Table of the switch again.



Main Diagram

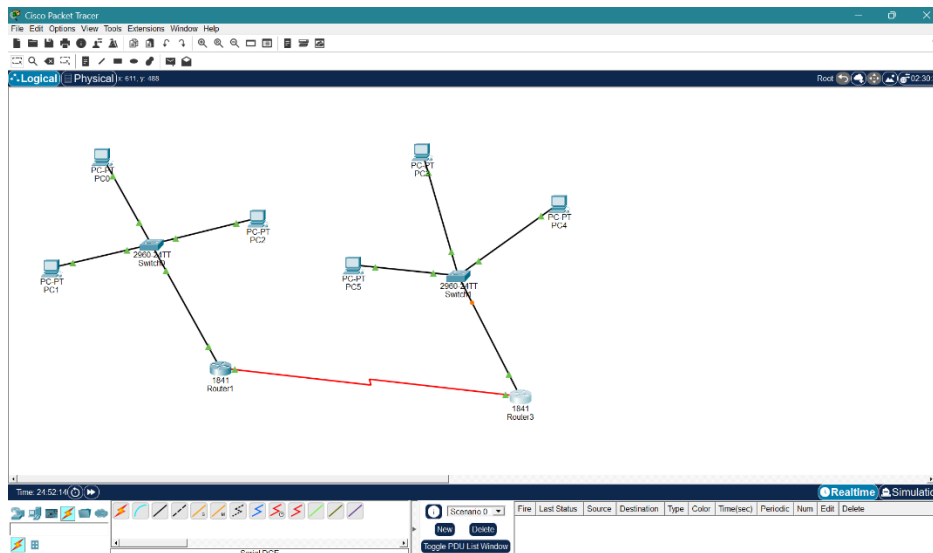


Empty MAC Table



PC0 PING

which is connected to the LAN. Connect the two routers through appropriate WAN interfaces. Assign IP addresses to the WAN interfaces from network 192.168.150.0/24. Add static route in both of the routers to route packets between two LANs.



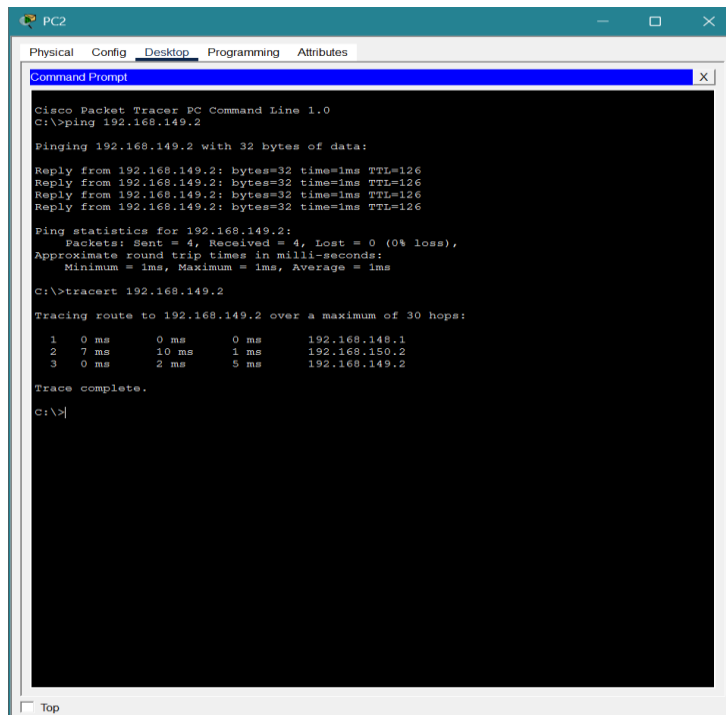
Main Diagram

```

PC0
Physical Config Desktop Programming Attributes
Command Prompt
Tracing route to 192.168.148.2 over a maximum of 30 hops:
 1  5 ms  2 ms  0 ms  192.168.148.2
Trace complete.
C:\>ping 192.168.148.1
Pinging 192.168.148.1 with 32 bytes of data:
Reply from 192.168.148.1: bytes=32 time<1ms TTL=255
Reply from 192.168.148.1: bytes=32 time<1ms TTL=255
Reply from 192.168.148.1: bytes=32 time<1ms TTL=255
Reply from 192.168.148.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.148.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.148.2
Pinging 192.168.148.2 with 32 bytes of data:
Reply from 192.168.148.2: bytes=32 time=6ms TTL=128
Reply from 192.168.148.2: bytes=32 time<1ms TTL=128
Reply from 192.168.148.2: bytes=32 time=8ms TTL=128
Reply from 192.168.148.2: bytes=32 time=8ms TTL=128
Ping statistics for 192.168.148.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 5ms
C:\>tracert 192.168.148.2
Tracing route to 192.168.148.2 over a maximum of 30 hops:
 1  6 ms  2 ms  9 ms  192.168.148.2
Trace complete.
C:\>tracert 192.168.148.3
Tracing route to 192.168.148.3 over a maximum of 30 hops:
 1  0 ms  1 ms  0 ms  192.168.148.3
Trace complete.

```

LAN 1 Internal Ping Check



PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.149.2

Pinging 192.168.149.2 with 32 bytes of data:

Reply from 192.168.149.2: bytes=32 time=1ms TTL=126
Reply from 192.168.149.2: bytes=32 time=1ms TTL=126
Reply from 192.168.149.2: bytes=32 time=1ms TTL=126
Reply from 192.168.149.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.149.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>tracert 192.168.149.2

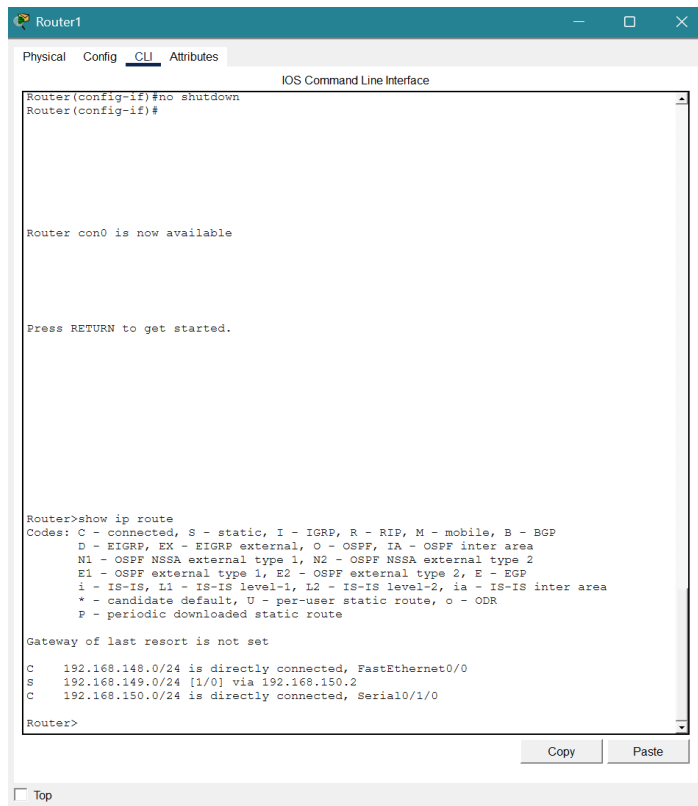
Tracing route to 192.168.149.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms   192.168.148.1
  1  7 ms    10 ms   1 ms   192.168.150.2
  2  0 ms    2 ms    5 ms   192.168.149.2

Trace complete.
C:\>
```

☐ Top

LAN1 to LAN2 Ping Check



Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#no shutdown
Router(config-if)#

Router con0 is now available

Press RETURN to get started.

Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.148.0/24 is directly connected, FastEthernet0/0
S    192.168.149.0/24 [1/0] via 192.168.150.2
C    192.168.150.0/24 is directly connected, Serial0/1/0

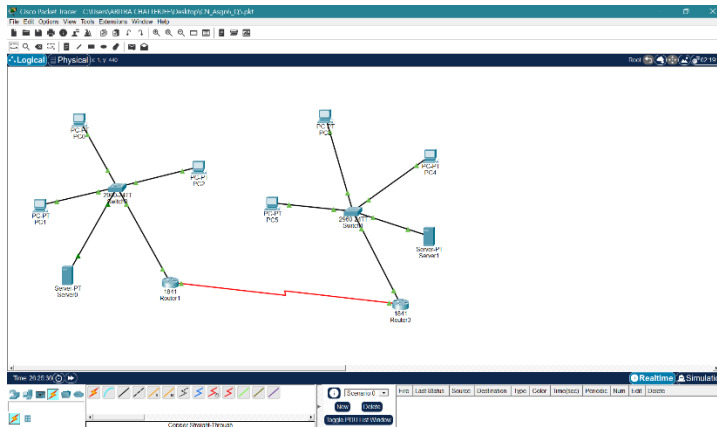
Router>
```

☐ Top

Copy Paste

Router Connection Proof

6. Add servers to the individual LANs (in problem 5) and configure them as a DHCPserver. Configure the hosts in the individual LAN to obtain IP addresses and address of the default gateway via this DHCP server.

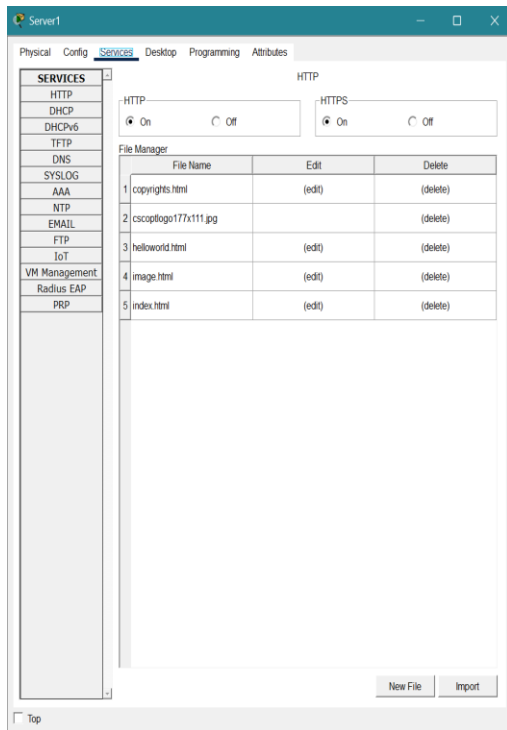


Main Diagram

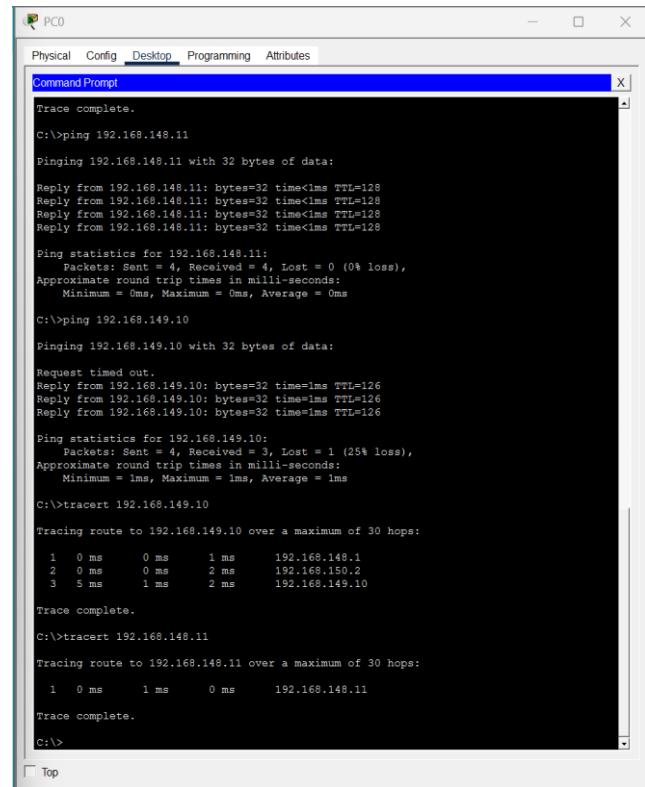
Server0 Configuration window showing the DHCP service configuration. The interface is FastEthernet0, and the service is turned on. The pool name is serverPool, and the default gateway is 192.168.148.1. The start IP address is 192.168.255.148, and the subnet mask is 255.255.255.0. The maximum number of users is 246. The TFTP server and WLC address are both 0.0.0.0.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.148.1	0.0.0.0	192.168.255.148	255.255.255.0	246	0.0.0.0	0.0.0.0

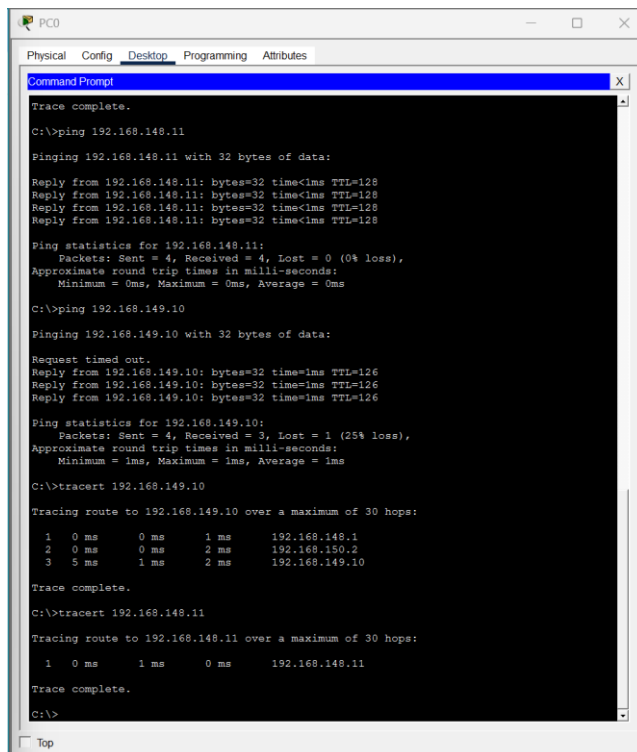
Server 0 Configure



PC0 DHCP

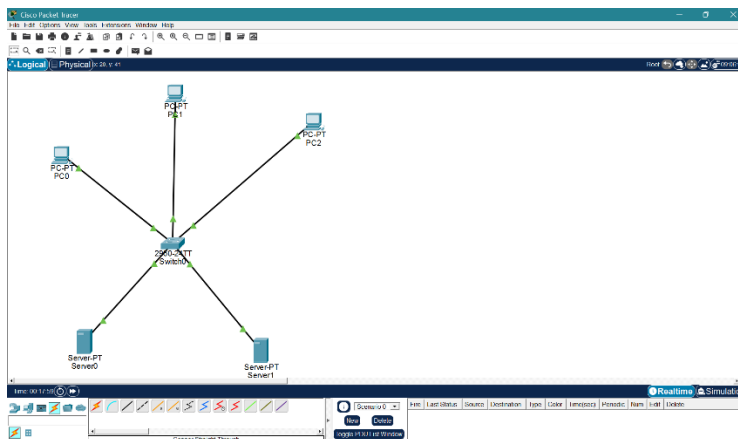


Ping from PC0

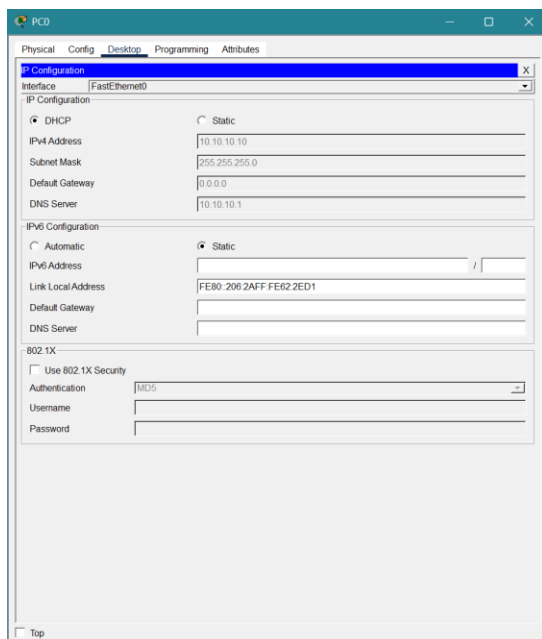


IP Config Summary PC1

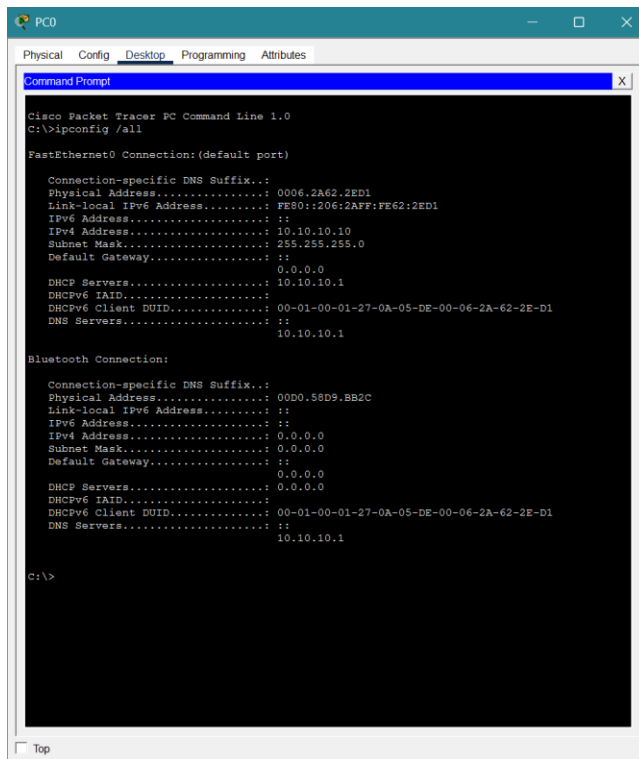
7. Create a LAN (CSE) with three hosts connected via a layer-2 switch (Cisco 2950 switch CSE-Switch). Also add a web server and a ftp server to this LAN. The hosts dynamically get their IP addresses from a local DHCP server. Servers are assigned fixed IP addresses. Configure the individual hosts to use the local DNS server for name resolution. Add a Domain Name Server (DNS) to this LAN. Create appropriate records in the DNS server for the individual servers in the LAN. The domain name of the LAN is cse.myuniv.edu. Configure the individual hosts to use the local DNS server for name resolution.



Main Diagram



PC0 DHCP



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

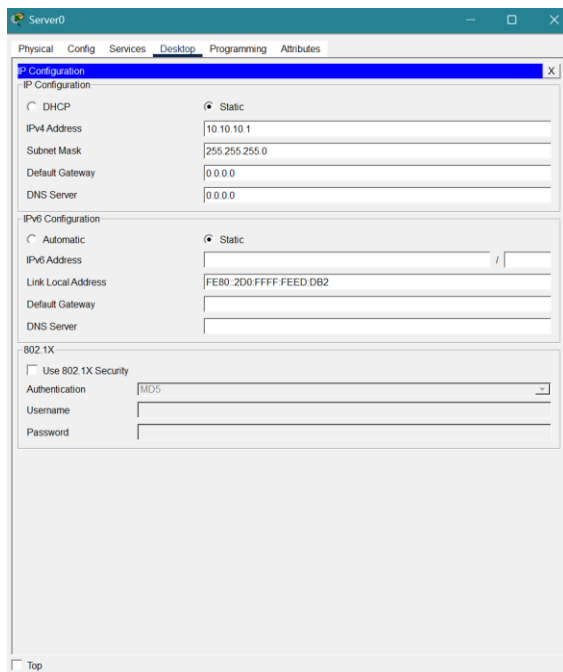
    Connection-specific DNS Suffix...: 
    Physical Address...: 0006.2A62.2ED1
    Link-local IPv6 Address...: FE80::206:2AFF:FE62:2ED1
    IPv6 Address...: ::
    IPv4 Address...: 10.10.10.10
    Subnet Mask...: 255.255.255.0
    Default Gateway...: ::
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...: 10.10.10.1
    DHCPv6 Client DUID...: 00-01-00-01-27-0A-05-DE-00-06-2A-62-2E-D1
    DNS Servers...: ::
    : 10.10.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address...: 00D0.58D9.BB2C
    Link-local IPv6 Address...: ::
    IPv6 Address...: ::
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: ::
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...: 0.0.0.0
    DHCPv6 Client DUID...: 00-01-00-01-27-0A-05-DE-00-06-2A-62-2E-D1
    DNS Servers...: ::
    : 10.10.10.1

C:\>
```

PC 0 Final IP Configure



Server0
Physical Config Services Desktop Programming Attributes

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.10.10.1

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2D0:FFFF:FEED:DB2

Default Gateway:

DNS Server:

802.1X

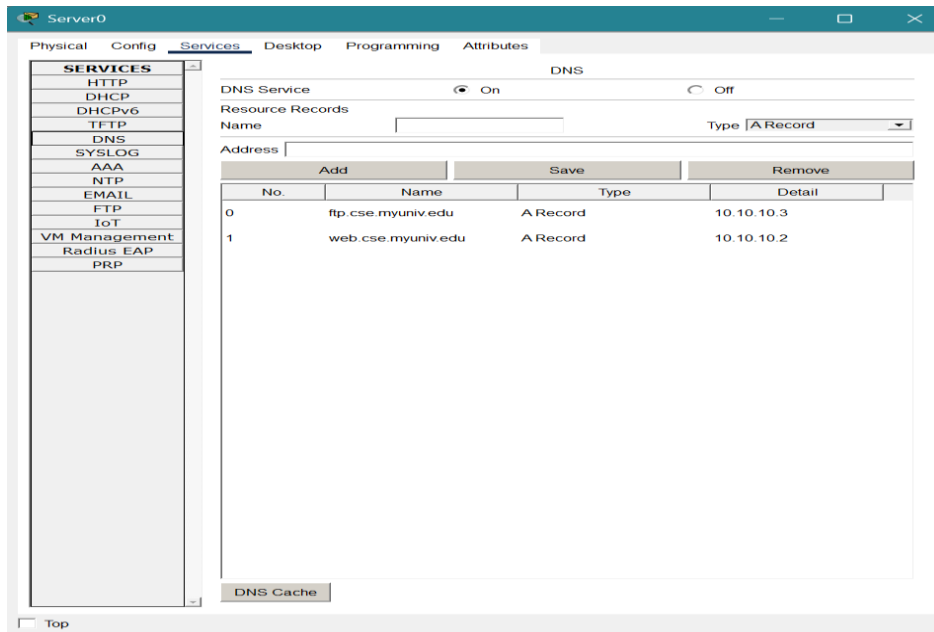
☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

DNS Server Config



DNS Table

Comments :

The CISCO packet tracer is actually a pretty useful tool in simulating the Network Layout. It can help to plan the architecture, simulate events and then deploy it in real life so that we can be aware about the performance of the network in production. Further it helps experience a practical knowledge of how elements of an entire work are when deployed in an integrated fashion. Hoping to get more of such assignments to that learning becomes fun and not stressful .